MATLAB

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Command Window

untitled1112.m untitled1111.m untitled123.m untitled2202.m untitled3433.m untitled004.m untitled345.m untitled2

/MATLAB Drive/untitled1112.m

```
clear; clc; close all;

Fs = 8000;
fc = 1500;
wc = fc / (Fs/2); % Normalized cutoff frequency
N = 50;          % Filter order

% Window Functions
w_rect  = rectwin(N+1);
w_hamm  = hamming(N+1);
w_hann  = hann(N+1);
w_black = blackman(N+1);

% FIR Filter Design using fir1
b_rect  = fir1(N, wc, w_rect);
b_hamm  = fir1(N, wc, w_hamm);
b_hann  = fir1(N, wc, w_hann);
b_black = fir1(N, wc, w_black);

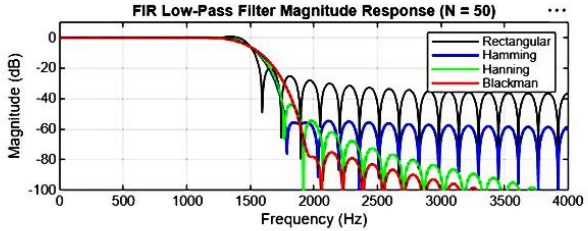
% Frequency Response Calculation
[H_rect, f] = freqz(b_rect, 1, 1024, Fs);
[H_hamm, ~] = freqz(b_hamm, 1, 1024, Fs);
[H_hann, ~] = freqz(b_hann, 1, 1024, Fs);
[H_black, ~] = freqz(b_black, 1, 1024, Fs);

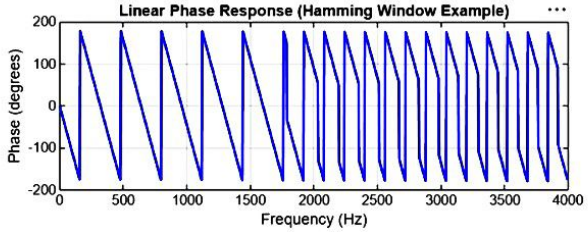
% ========================================
% DISPLAY NUMERICAL OUTPUTS IN COMMAND WINDOW
% ========================================
fprintf('=== EXPERIMENT 4: FIR WINDOWING TECHNIQUES ===\n\n');
fprintf('Filter Order (N): %d\n', N);
fprintf('Cutoff Frequency: %d Hz\n\n', fc);

fprintf('--- RECTANGULAR WINDOW COEFFICIENTS (First 5) ---\n');
disp(b_rect(1:5));

fprintf('--- HAMMING WINDOW COEFFICIENTS (First 5) ---\n');
disp(b_hamm(1:5));
```

Figure 1: FIR Windowing Techniques Comparison





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